# Junwoo Chang

GitHub | Homepage | Google Scholar | junwoo_chang@berkeley.edu

## Research Interests

Robotics, Reinforcement Learning, Geometric Deep Learning

## Education

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|---|---|
| Aug. 2026 – | Incoming Ph.D. student, **UC Berkeley** (Advisor: Prof. Negar Mehr) |
| Mar. 2024 – Feb. 2026 | M.S., **Yonsei University** (Advisor: Prof. Jongeun Choi) |
| Mar. 2018 – Feb. 2024 | B.S., **Yonsei University**<br>Leave of absence for mandatory military service (Jul. 2019 – Jan. 2021) |

## Publications

(*: equal contribution, †: equal advising)

- Minwoo Park*, **Junwoo Chang***, Jongeun Choi, Roberto Horowitz
  Symmetry-Aware Steering of Equivariant Diffusion Policies: Benefits and Limits
  International Federation of Automatic Control **(IFAC), 2026**

- Jebeom Chae*, **Junwoo Chang***, Seungho Yeom, Yujin Kim, Jongeun Choi
  Multi-Robot Motion Planning from Vision and Language using Heat-Inspired Diffusion
  IEEE Robotics and Automation Letters **(RA-L), 2026**

- **Junwoo Chang**, Minwoo Park, Joohwan Seo, Roberto Horowitz, Jongmin Lee†, Jongeun Choi†
  Partially Equivariant Reinforcement Learning in Symmetry-Breaking Environments
  *International Conference on Learning Representations* **(ICLR), 2026**

- **Junwoo Chang**, Joseph Park, Roberto Horowitz, Jongmin Lee†, Jongeun Choi†
  Group-Invariant Unsupervised Skill Discovery: Symmetry-aware Skill Representations for Generalizable Behavior
  *Preprint* (under review), 2026

- Joohwan Seo, Soochul Yoo, **Junwoo Chang**, Hyunseok An, Hyunwoo Ryu, Soomi Lee, Arvind Kruthiventy, Jongeun Choi, Roberto Horowitz
  SE(3)-Equivariant Robot Learning and Control: A Tutorial Survey
  *International Journal of Control, Automation and Systems* **(IJCAS), 2025**

- Hyunwoo Ryu, Jiwoo Kim, Hyunseok An, **Junwoo Chang**, Joohwan Seo, Taehan Kim, Yubin Kim, Chaewon Hwang, Jongeun Choi, Roberto Horowitz
  Diffusion-EDFs: Bi-equivariant Denoising Generative Modeling on SE(3) for Visual Robotic Manipulation
  *Computer Vision and Pattern Recognition* **(CVPR), 2024 (Highlight)**

- **Junwoo Chang***, Hyunwoo Ryu*, Jiwoo Kim, Soochul Yoo, Joohwan Seo, Nikhil Potu Surya Prakash, Jongeun Choi, Roberto Horowitz
  Denoising Heat-inspired Diffusion with Insulators for Collision Free Motion Planning
  **NeurIPS 2023 Workshop on Diffusion Models**

## Awards and Scholarships

**Best Technical Presentation Award** Oct. 2023
The 5th Yonsei University Mechanical Engineering Graduate Student Academic Conference

**Yonsei Jinri Scholarship** Dec. 2021, Jul. 2022, Dec. 2022
Recognized for sustained academic excellence (three consecutive awards)

**Undergraduate Academic Excellence Honors** 2021
High Honors (Spring 2021), Honors (Fall 2021)

## Research Experience

**Machine Learning and Control Systems Laboratory**, Yonsei University Mar. 2024 – Present
*Graduate Researcher, Advisor: Prof. Jongeun Choi*
– Research on group equivariant and diffusion-based robot learning
– Developing partially equivariant reinforcement learning methods for symmetry-breaking tasks

**Machine Learning and Control Systems Laboratory**, Yonsei University Sep. 2022 – Feb. 2024
*Undergraduate Research Intern, Advisor: Prof. Jongeun Choi*
– **Hyundai Motor Project:** Self-supervised representation learning for autonomous driving
– Integrated heat-transfer dynamics with diffusion models for vision-based navigation
– **Undergraduate Thesis:** *Imaginary Experience Replay: Generating Redundant Transitions for Sparse and Negative Rewards*

**Human–Centered AI Robotics Laboratory**, Yonsei University Jun. 2022 – Sep. 2022
*Undergraduate Research Intern, Advisor: Prof. Dongjun Shin*
– Designed a 5-DOF snake robot for in-pipe locomotion and haptic teleoperation

## Project Experience

**Technical Demonstration of Diffusion-EDFs** Aug. 2023 – Oct. 2023
– Demonstrated real-world robotic manipulation using the Diffusion-EDFs
– Awarded Best Technical Demonstration (top project at conference)

**Volunteer Research, Yonsei Rehabilitation Hospital** Jun. 2022 – Jan. 2023
– Designed assistive systems for children with mental and physical disabilities
– Developed a collision-aware wheelchair assistant and a posture correction aid for children

## Teaching

**Teaching Assistant**
*Mechanical Engineering Laboratory II (Yonsei MEU3005-01), Prof. Jongeun Choi* Spring 2024

## Academic Service

– Reviewer, Neural Information Processing Systems (NeurIPS) 2026
– Reviewer, European Conference on Computer Vision (ECCV) 2026
– Reviewer, IEEE International Conference on Robotics and Automation (ICRA) 2026
– Reviewer, IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2026
– Reviewer, IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM) 2024

## Military Service

**Republic of Korea Army**, K-SAM Pegasus Air Defense System Operator Jul. 2019 – Jan. 2021

## Skills

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| **Programming** | C, C++, Python (PyTorch, JAX, TensorFlow), MATLAB, ROS |
| **Hardware** | Franka Emika, Kinova Gen2, RB-Y1, TurtleBot3, OptiTrack, Arduino, Raspberry Pi |
| **Tools** | MoveIt, Git, Linux, LaTeX |
| **Theory** | Reinforcement Learning, Representation Theory, Group Theory, Diffusion Models |